AP Macroeconomics Lecture Notes

1. Introduction to Macroeconomics

Macroeconomics studies the economy as a whole rather than individual consumers or firms.

Focuses on:

- Economic growth
- Inflation
- Unemployment
- Recessions
- Government policy
- International trade

Key macro questions:

- Why do recessions happen?
- Why do prices rise?
- Why are some countries richer than others?
- How can governments stabilize the economy?

2. Gross Domestic Product (GDP)

Definition

GDP = total market value of all **final goods and services** produced **within a country** in a given time period.

Keywords:

- **market value** → measured in dollars
- **final goods** → avoids double counting
- **within a country** → domestic production
- **time period** → usually quarterly or yearly

GDP Formula

$GDP = C + I + G + (X - M)$

Where:

C = Consumption

Household spending:

- food
- clothes
- phones
- rent
- healthcare

Largest GDP component.

I = Investment

Business spending on:

- factories
- machinery
- software
- buildings

Also includes:

- inventory changes
- new residential construction

NOT stock purchases.

G = Government Spending

Government purchases:

- roads
- military
- schools

- public salaries

Does NOT include:

- Social Security
- welfare
- unemployment benefits

(these are transfer payments)

X − M = Net Exports

Exports = goods sold abroad

Imports = foreign goods bought domestically

Net exports:

If exports > imports:
Trade surplus

If imports > exports:
Trade deficit

3. Nominal vs Real GDP

Nominal GDP

Measured using current prices.

Inflation affects it.

Example:
2025 phone = $1000
2026 same phone = $1200

Nominal GDP rises even if output unchanged.

Real GDP

Adjusted for inflation.

Uses base-year prices.

Measures actual production changes.

Better measure of economic growth.

4. GDP Limitations

GDP does NOT measure:

- happiness
- pollution
- income inequality
- unpaid work
- black market activity
- quality improvements
- leisure time

Example:
A parent caring for a child at home = not counted

Hiring babysitter = counted

5. Business Cycle

Economies move in cycles.

Stages:

Expansion

Economy grows:

- rising output
- more jobs
- higher income

Peak

Highest point before slowdown.

Contraction / Recession

Economic decline:

- lower spending
- layoffs
- reduced production

Trough

Lowest point before recovery.

6. Unemployment

Formula

Unemployment Rate =
(Unemployed / Labor Force) × 100

Labor force =
employed + unemployed actively seeking work

Types of Unemployment

Frictional

Temporary unemployment while switching jobs.

Example:
college graduate job hunting

Normal.

Structural

Skills mismatch.

Example:
factory worker replaced by automation

Harder to fix.

Cyclical

Caused by recession.

Example:
restaurants lay off workers due to low demand

Most dangerous during downturns.

Seasonal

Jobs tied to time of year.

Examples:

- ski resorts
- holiday retail

7. Natural Rate of Unemployment

Includes:

- frictional
- structural

Excludes:

- cyclical

Economy can never realistically have 0% unemployment.

8. Inflation

Definition

General rise in price levels over time.

Reduces purchasing power.

Example:
$10 buys less than before.

Measuring Inflation

Consumer Price Index (CPI)

Tracks price changes in a basket of consumer goods.

Examples:

- groceries
- gas
- rent
- transportation

Inflation Formula

Inflation Rate =
[(New CPI − Old CPI) / Old CPI] × 100

9. Causes of Inflation

Demand-Pull Inflation

“Too much money chasing too few goods”

Demand rises faster than supply.

Example:
stimulus checks increase spending.

Cost-Push Inflation

Production costs rise.

Businesses raise prices.

Examples:

- oil shocks
- wage increases
- supply shortages

Monetary Inflation

Too much money supply growth.

10. Effects of Inflation

Hurts:

- savers
- fixed-income earners
- lenders

Helps:

- borrowers

Because debt becomes easier to repay.

11. Aggregate Demand (AD)

Total spending in economy.

Formula:

$AD = C + I + G + (X - M)$

Downward sloping because:

Wealth Effect

Lower prices → consumers feel richer → spend more

Interest Rate Effect

Lower prices → lower interest rates → more borrowing

Net Export Effect

Lower domestic prices → exports rise

12. Aggregate Supply (AS)

Short-Run Aggregate Supply (SRAS)

Production when wages/prices sticky.

Upward sloping.

Higher prices → firms produce more.

Long-Run Aggregate Supply (LRAS)

Economy’s full employment output.

Vertical.

Determined by:

- labor
- capital
- technology
- productivity

13. AD/AS Changes

AD increases

Causes:

- consumer confidence
- tax cuts
- government spending

- lower rates

Results:

- higher output
- higher prices

AD decreases

Results:

- recession
- lower output
- unemployment

SRAS decreases

Causes:

- oil price spikes
- supply shocks

Results:

- inflation
- lower output

This is stagflation.

14. Stagflation

Combination of:

- inflation
- unemployment
- slow growth

Usually caused by supply shocks.

Classic example:
1970s oil crisis

15. Fiscal Policy

Government adjusts:

- spending
- taxes

Expansionary Fiscal Policy

Used during recession.

Government:

- increases spending
  OR
- cuts taxes

Goal:
increase AD

Contractionary Fiscal Policy

Used during inflation.

Government:

- cuts spending
- raises taxes

Goal:
reduce AD

16. Multiplier Effect

Initial spending creates larger total economic impact.

Formula:

Multiplier =
1 / (1 − MPC)

MPC = marginal propensity to consume

Example:
MPC = 0.8

Multiplier:
1 / (1 − 0.8) = 5

$100 spending creates $500 total impact.

17. Crowding Out

Government borrowing raises interest rates.

Higher rates reduce private investment.

Bad side effect of fiscal expansion.

18. National Debt vs Budget Deficit

Budget deficit:
one-year shortfall

Debt:
total accumulated deficits

19. Money

Functions:

Medium of Exchange

Used to buy goods

Store of Value

Keeps value over time

Unit of Account

Measures prices

20. Federal Reserve

Central bank of U.S.

Controls money supply.

Goals:

- stable prices
- full employment
- economic growth

21. Monetary Policy Tools

Open Market Operations

Fed buys/sells bonds

BUY bonds:
inject money
lower rates

SELL bonds:
remove money
raise rates

Reserve Requirement

Amount banks must keep.

Lower reserve:
more lending

Higher reserve:
less lending

Discount Rate

Interest rate charged to banks.

Lower:
encourages borrowing

Higher:
discourages borrowing

22. Expansionary Monetary Policy

Used during recession.

Fed:

- buys bonds
- lowers discount rate
- lowers reserve requirement

Result:
more money
lower interest
more spending

23. Contractionary Monetary Policy

Used during inflation.

Fed:

- sells bonds
- raises rates
- raises reserve requirements

Result:
less spending

24. Fractional Reserve Banking

Banks keep only part of deposits.

Lend the rest.

Creates money through repeated lending.

25. Money Multiplier

Formula:

1 / Reserve Ratio

Example:
10% reserve ratio

1 / 0.10 = 10

$1000 deposit → $10,000 potential money creation

26. Loanable Funds Market

Supply:
saving

Demand:
borrowing

Interest rate determined here.

If government borrowing rises:
Demand shifts right

Interest rates rise.

27. Foreign Exchange

Currency market.

Demand for currency rises when foreigners want domestic goods.

Appreciation

Currency stronger

Effects:

- imports cheaper
- exports more expensive

Depreciation

Currency weaker

Effects:

- exports cheaper
- imports more expensive

28. Comparative Advantage

Countries should specialize where opportunity cost is lower.

This increases efficiency and trade gains.

29. Economic Growth

Long-run growth depends on:

- education
- technology
- capital
- productivity

- infrastructure
- stable institutions

30. Phillips Curve

Shows relationship between:
inflation and unemployment

Short run:
inverse relationship

Long run:
vertical at natural unemployment rate

AP Exam FRQ Strategy

For graph questions:

Always:

1. identify curve
2. say shift direction
3. explain WHY
4. state outcome

Example:
“AD shifts right because government spending increased, raising output and price level.”